7 **(a)** In the magnetic field, the magnetic force acting on the ion provides the centripetal force for the ion to move in uniform circular motion. Thus **B1**

$$Bqv = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{Bq}$$

Since $T = \dfrac{2\pi r}{v}$,

$$\therefore T = \frac{2\pi}{v}\left(\frac{mv}{Bq}\right) \Rightarrow T = \frac{2\pi m}{Bq}$$

which is independent of *r*. **B1**

Assume that the time taken by the ion to cross the gaps is negligible compared to the time taken by the ion to travel in the magnetic field.

B1

(b)(i) **C1**

$$\therefore T = \frac{2\pi(6.68 \text{ x } 10^{-27})}{0.85 \text{ x } 2 \text{ x } 1.6 \text{ x } 10^{-19}}$$ **A1**

$$= 1.54 \text{ x } 10^{-7} \text{ s}$$

(b)(ii) In order for the nucleus to accelerate when it crosses the gap, freq. of the alternating voltage = orbital freq. of the nucleus **B1**

$$\therefore f = \frac{1}{1.54 \text{ x } 10^{-7}} = 6.49 \text{ x } 10^{6} \text{ Hz}$$

A1

(b)(iii) The **work done by the magnetic force on the ion is zero** since the **magnetic force is always perpendicular to the velocity of the ion**. **A1**

Each time the ion crosses the gap, it **gains a kinetic energy of *qV*** due to the work done on it by the electric field given by $Fd = qEd = q(V/d)d$ where F is the electric force acting on the ion, d is the separation between the gaps and E is the strength of the uniform electric field between the gaps. In **one revolution**, the ion will cross the gap **two times**. **A1**

Thus, the total gain in its kinetic energy is $2qV = 4eV$, since the charge of the helium nucleus is $2e$.

8 (a) A longitudinal wave is one in which the oscillation of the molecules of the